

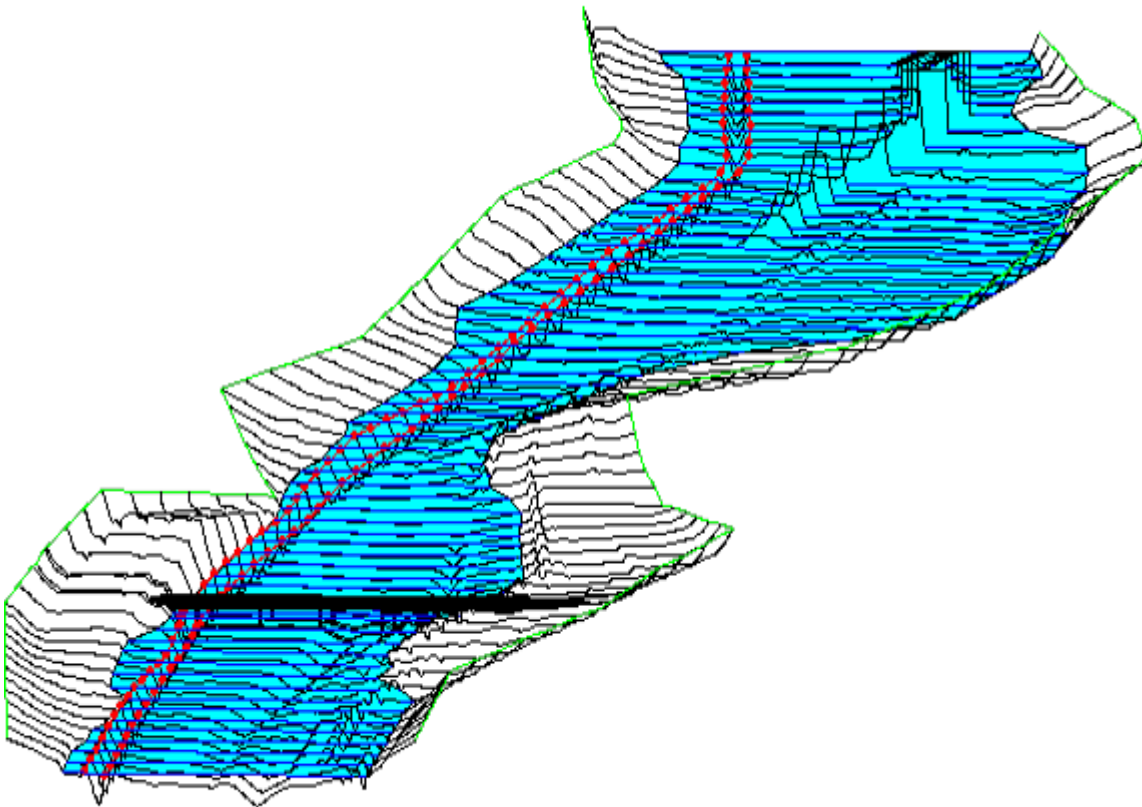


**US Army Corps
of Engineers®**

Hydrologic Engineering Center

HEC-RAS

River Analysis System



1D Unsteady Flow Hydrodynamics

Version 6.5

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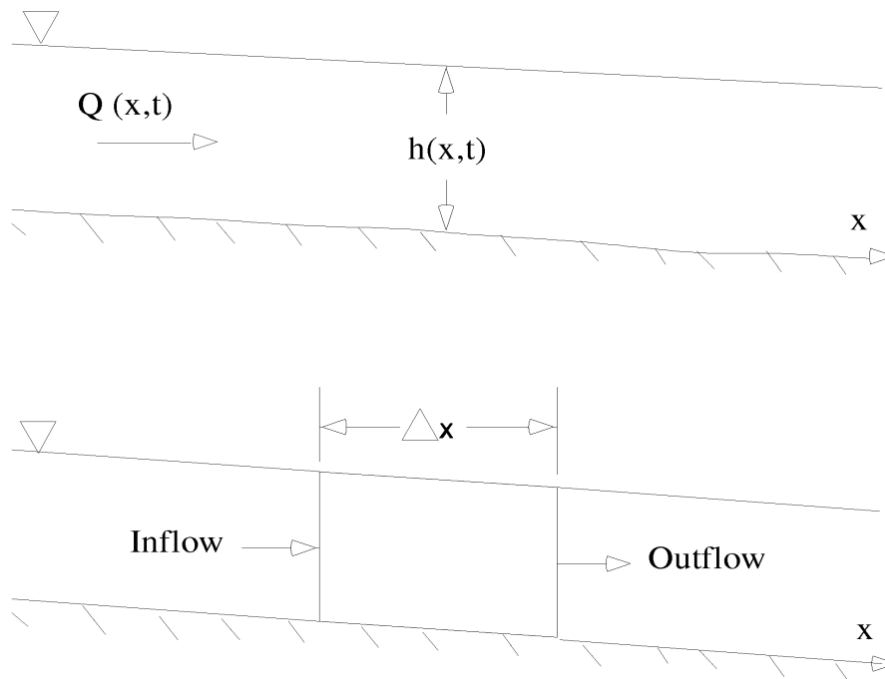
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The physical laws which govern the flow of water in a stream are: (1) the principle of conservation of mass (continuity), and (2) the principle of conservation of momentum. These laws are expressed mathematically in the form of partial differential equations, which will hereafter be referred to as the continuity and momentum equations. The derivations of these equations are presented in this chapter based on a paper by James A. Liggett from the book *Unsteady Flow in Open Channels* (Mahmmod and Yevjevich, 1975).

1 CONTINUITY EQUATION

Consider the elementary control volume shown in the figure below. In this figure, distance x is measured along the channel, as shown. At the midpoint of the control volume the flow and total flow area are denoted $Q(x,t)$ and A_T , respectively. The total flow area is the sum of active area A and off-channel storage area S .



Conservation of mass for a control volume states that the net rate of flow into the volume be equal to the rate of change of storage inside the volume. The rate of inflow to the control volume may be written as:

$$1) \quad Q - \frac{\partial Q}{\partial x} \frac{\Delta x}{2}$$

the rate of outflow as:

$$2) \quad Q + \frac{\partial Q}{\partial x} \frac{\Delta x}{2}$$

and the rate of change in storage as:

$$3) \quad \frac{\partial A_T}{\partial t} \Delta x$$

Assuming that Δx is small, the change in mass in the control volume is equal to:

$$4) \quad \rho \frac{\partial A_T}{\partial t} \Delta x = \rho \left[\left(Q - \frac{\partial Q}{\partial x} \frac{\Delta x}{2} \right) - \left(Q + \frac{\partial Q}{\partial x} \frac{\Delta x}{2} \right) + Q_i \right]$$

Symbol	Description	Units
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Q_i is the lateral flow entering the control volume and ρ is the fluid density.

Simplifying and dividing through by ρ yields the final form of the continuity equation:

$$5) \quad \frac{\partial A_T}{\partial t} + \frac{\partial Q}{\partial x} = q_t$$

Symbol	Description	Units
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q_t lateral inflow per unit length

2 MOMENTUM EQUATION

Conservation of momentum is expressed by Newton's second law as:

$$6) \quad \sum F_x = \frac{d\vec{M}}{dt}$$

Conservation of momentum for a control volume states that *the net rate of momentum entering the volume (momentum flux) plus the sum of all external forces acting on the volume be equal to the rate of accumulation of momentum*. This is a vector equation applied in the x-direction. The momentum flux (MV) is the fluid mass times the velocity vector in the direction of flow. Three forces will be considered: (1) pressure, (2) gravity and (3) boundary drag, or friction force.

Pressure forces: The figure below illustrates the general case of an irregular cross section. The pressure distribution is assumed to be hydrostatic (pressure varies linearly with depth) and the total pressure force is the integral of the pressure-area product over the cross section. After Shames (1962), the pressure force at any point may be written as:

$$7) \quad F_p = \int_0^h \rho g(h-y)T(y)dy$$

Symbol	Description	Units
h	depth	
y	distance above the channel invert	
$T(y)$	width function which relates the cross section width to the distance above the channel invert	

If F_p is the pressure force in the x-direction at the midpoint of the control volume, the force at the upstream end of the control volume may be written as

$$8) \quad F_p - \frac{\partial F_p}{\partial x} \frac{\Delta x}{2}$$

and at the downstream end as:

$$9) \quad F_p + \frac{\partial F_p}{\partial x} \frac{\Delta x}{2}$$

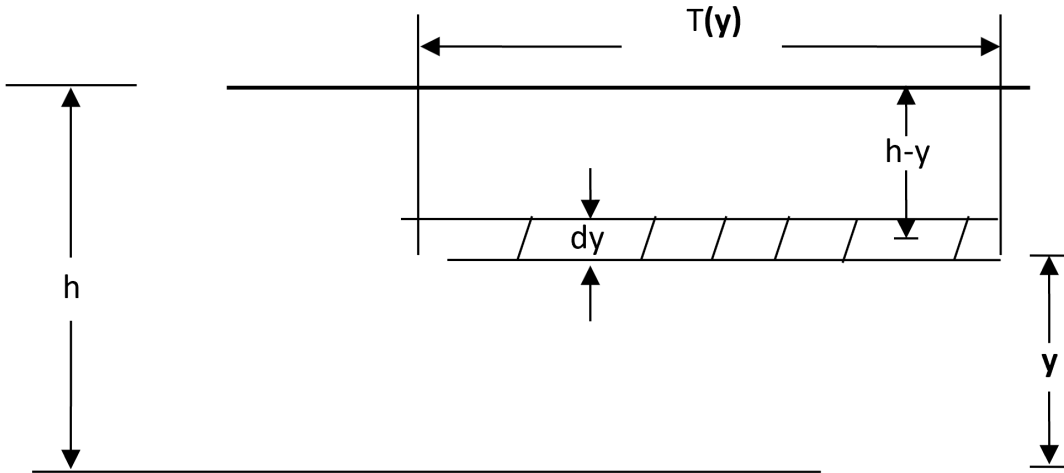


Illustration of Terms Associated with Definition of Pressure Force

The sum of the pressure forces for the control volume may therefore be written as:

$$10) \quad F_{Pn} = \left| F_P - \frac{\partial F_P}{\partial x} \frac{\Delta x}{2} \right| - \left| F_P + \frac{\partial F_P}{\partial x} \frac{\Delta x}{2} \right| + F_B$$

Where F_{Pn} is the net pressure force for the control volume, and F_B is the force exerted by the banks in the x-direction on the fluid. This may be simplified to:

$$11) \quad F_{Pn} = -\frac{\partial F_P}{\partial x} \Delta x + F_B$$

Differentiating (7) using Leibnitz's Rule and then substituting in (11) results in:

$$12) \quad F_{Pn} = -\rho g \Delta x \left[\frac{\partial h}{\partial x} \int_0^h T(y) dy + \int_0^h (h-y) \frac{\partial T(y)}{\partial x} dy \right] + F_B$$

The first integral in (12) is the cross-sectional area, A . The second integral (multiplied by $-\rho g \Delta x$) is the pressure force exerted by the fluid on the banks, which is exactly equal in magnitude, but opposite in direction to F_B . Hence the net pressure force may be written as:

$$13) \quad F_{Pn} = -\rho g A \frac{\partial h}{\partial x} \Delta x$$

Gravitational force: The force due to gravity on the fluid in the control volume in the x-direction is:

$$14) \quad F_g = \rho g A \sin \theta \Delta x$$

here θ is the angle that the channel invert makes with the horizontal. For natural rivers θ is small and $\sin\theta \approx \tan\theta = -\frac{\delta z_0}{\delta X}$, where z_0 is the invert elevation. Therefore the gravitational force may be written as

$$15) \quad F_g = -\rho g A \frac{\partial z_0}{\partial x} \Delta x$$

This force will be positive for negative bed slopes.

Boundary drag (friction force): Frictional forces between the channel and the fluid may be written as:

$$16) \quad F_f = -\tau_0 P \Delta x$$

where τ_0 is the average boundary shear stress (force/unit area) acting on the fluid boundaries, and P is the wetted perimeter. The negative sign indicates that, with flow in the positive x-direction, the force acts in the negative x-direction. From dimensional analysis, τ_0 may be expressed in terms of a drag coefficient, C_D , as follows:

$$17) \quad \tau_0 = \rho C_D V^2$$

The drag coefficient may be related to the Chezy coefficient, C , by the following:

$$18) \quad C_D = \frac{g}{C^2}$$

Further, the Chezy equation may be written as:

$$19) \quad V = C \sqrt{RS_f}$$

Substituting (17), (18), and (19) into (16), and simplifying, yields the following expression for the boundary drag force:

$$20) \quad F_f = -\rho g A S_f \Delta x$$

where S_f is the friction slope, which is positive for flow in the positive x-direction. The friction slope must be related to flow and stage. Traditionally, the Manning and Chezy friction equations have been used. Since the Manning equation is predominantly used in the United States, it is also used in HEC-RAS. The Manning equation is written as:

$$21) \quad S_f = \frac{Q |Q| n^2}{2.208 R^{4/3} A^2}$$

Symbol	Description	Units
R	hydraulic radius and n is the Manning friction coefficient	

Momentum flux: With the three force terms defined, only the momentum flux remains. The flux entering the control volume may be written as:

$$22) \quad \rho \left[QV - \frac{\partial QV}{\partial x} \frac{\Delta x}{2} \right]$$

and the flux leaving the volume may be written as:

$$23) \quad \rho \left[QV + \frac{\partial QV}{\partial x} \frac{\Delta x}{2} \right]$$

Therefore the net rate of momentum (momentum flux) entering the control volume is:

$$24) \quad -\rho \frac{\partial(QV)}{\partial x} \Delta x$$

Since the momentum of the fluid in the control volume is $\rho Q \Delta x$, the rate of accumulation of momentum may be written as:

$$25) \quad \frac{\partial}{\partial t}(\rho Q \Delta x) = \rho \Delta x \frac{\partial Q}{\partial t}$$

Restating the principle of conservation of momentum:

The net rate of momentum (momentum flux) entering the volume (24) plus the sum of all external forces acting on the volume [(13) + (15) + (20)] is equal to the rate of accumulation of momentum (25). Hence:

$$26) \quad \rho \Delta x \frac{\partial Q}{\partial t} = -\rho \frac{\partial(QV)}{\partial x} \Delta x - \rho g A \frac{\partial h}{\partial x} \Delta x - \rho g A \frac{\partial z_0}{\partial x} \Delta x - \rho g A S_f \Delta x$$

The elevation of the water surface, z , is equal to $z_0 + h$. Therefore:

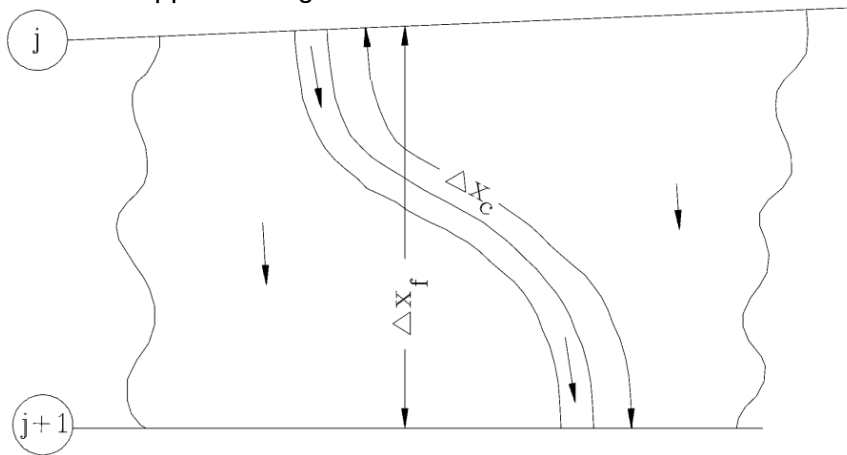
$$27) \quad \frac{\partial z_z}{\partial x} = \frac{\partial h}{\partial x} + \frac{\partial z_0}{\partial x}$$

where $\partial z/\partial x$ is the water surface slope. Substituting (27) into (26), dividing through by $\rho\Delta x$ and moving all terms to the left yields the final form of the momentum equation:

$$28) \quad \frac{\partial Q}{\partial t} + \frac{\partial(QV)}{\partial x} + gA \left(\frac{\partial z_s}{\partial x} + S_f \right) = 0$$

3 APPLICATION OF THE 1D UNSTEADY FLOW EQUATIONS WITHIN HEC-RAS

The figure below illustrates the two-dimensional characteristics of the interaction between the channel and floodplain flows. When the river is rising water moves laterally away from the channel, inundating the floodplain and filling available storage areas. As the depth increases, the floodplain begins to convey water downstream generally along a shorter path than that of the main channel. When the river stage is falling, water moves toward the channel from the overbank supplementing the flow in the main channel.



Because the primary direction of flow is oriented along the channel, this two-dimensional flow field can often be accurately approximated by a one-dimensional representation. Off-channel ponding areas can be modeled with storage areas that exchange water with the channel. Flow in the overbank can be approximated as flow through a separate channel.

This channel/floodplain problem has been addressed in many different ways. A common approach is to ignore overbank conveyance entirely, assuming that the overbank is used only for storage. This assumption may be suitable for large streams such as the Mississippi River where the channel is confined by levees and the remaining floodplain is either heavily vegetated or an off-channel storage area. Fread (1976) and Smith (1978) approached this problem by dividing the system into two separate channels and writing continuity and momentum equations for each channel. To simplify the problem they assumed a horizontal water surface at each cross section normal to the direction of flow; such that the exchange of momentum between the channel and the floodplain was negligible and that the discharge was distributed according to conveyance, i.e.:

$$29) \quad Q_c = \phi Q$$

Symbol	Description	Units
Q_c	flow in channel	
Q	total flow	

Symbol	Description	Units
ϕ	$\frac{K_c}{K_c + K_f}$	
K_c	conveyance in the channel	
K_f	conveyance in the floodplain	

With these assumptions, the one-dimensional equations of motion can be combined into a single set:

$$30) \quad \frac{\partial A}{\partial t} + \frac{\partial(\Phi Q)}{\partial x_c} + \frac{\partial[(1 - \Phi)Q]}{\partial x_f} = 0$$

$$31) \quad \frac{\partial Q}{\partial t} + \frac{\partial\left(\frac{\Phi^2 Q^2}{A_c}\right)}{\partial x_c} + \frac{\partial\left(\frac{(1-\Phi^2)Q^2}{A_f}\right)}{\partial x_f} + gA_c \left[\frac{\partial Z}{\partial x_c} + S_{fc} \right] + gA_f \left[\frac{\partial z}{\partial x_f} + S_{ff} \right] = 0$$

in which the subscripts c and f refer to the channel and floodplain, respectively. These equations were approximated using implicit finite differences, and solved numerically using the Newton-Raphson iteration technique. The model was successful and produced the desired effects in test problems. Numerical oscillations, however, can occur when the flow at one node, bounding a finite difference cell, is within banks and the flow at the other node is not.

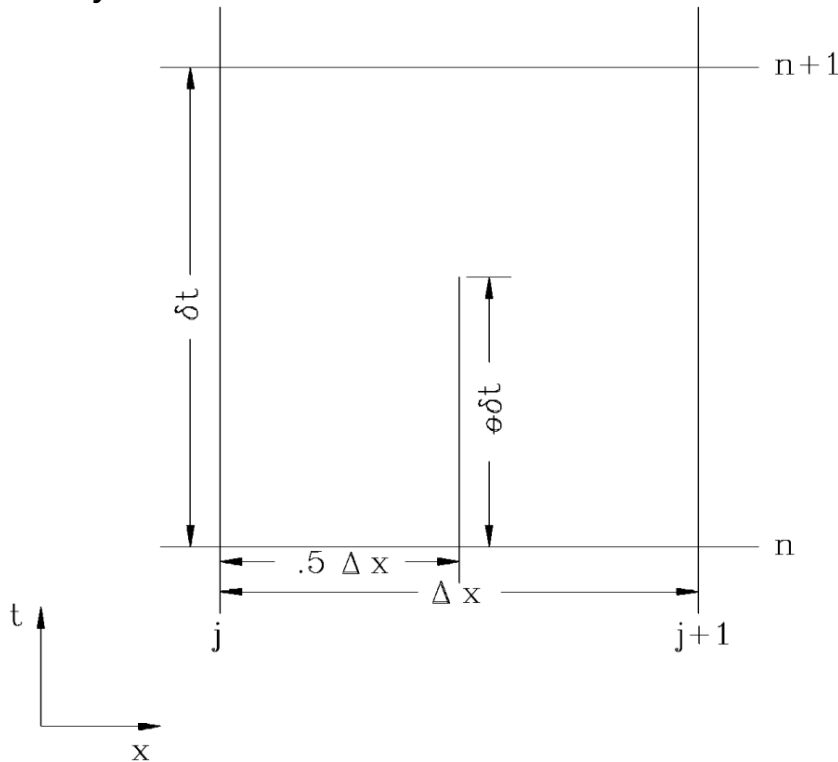
Expanding on the earlier work of Fread and Smith, Barkau (1982) manipulated the finite difference equations for the channel and floodplain and defined a new set of equations that were computationally more convenient. Using a velocity distribution factor, he combined the convective terms. Further, by defining an equivalent flow path, Barkau replaced the friction slope terms with an equivalent force.

The equations derived by Barkau are the basis for the unsteady flow solution within the HEC-RAS software. These equations were derived above. The numerical solution of these equations is described in the next sections.

4 IMPLICIT FINITE DIFFERENCE SCHEME

The most successful and accepted procedure for solving the one-dimensional unsteady flow equations is the four-point implicit scheme, also known as the box scheme (see figure below). Under this scheme, space derivatives and function values are evaluated at an interior point, $(n + \theta)\Delta t$. Thus values at $(n + 1)\Delta t$ enter into all terms in the equations. For a reach of river, a system of simultaneous equations results. The simultaneous solution is an important aspect of this scheme because it allows information from the entire reach to influence the solution at any one point. Consequently, the time step can be significantly larger than with explicit numerical schemes. Von Neumann stability analyses performed by Fread (1974), and Liggett and Cunge (1975), show the implicit scheme to be unconditionally stable (theoretically) for $0.5 < \theta \leq 1.0$, conditionally stable for $\theta = 0.5$, and unstable for $\theta < 0.5$. In a convergence analysis performed by the same authors, it was shown that numerical damping increased as the ratio $\lambda/\Delta x$ decreased, where λ is the length of a wave in the hydraulic system. For streamflow routing problems where the wavelengths are long with respect to spatial distances, convergence is not a serious problem.

In practice, other factors may also contribute to the non-stability of the solution scheme. These factors include dramatic changes in channel cross-sectional properties, abrupt changes in channel slope, characteristics of the flood wave itself, and complex hydraulic structures such as levees, bridges, culverts, weirs, and spillways. In fact, these other factors often overwhelm any stability considerations associated with θ . **Because of these factors, any model application should be accompanied by a sensitivity study, where the accuracy and the stability of the solution are tested with various time and distance intervals.**



The following notation is defined:

$$32) \quad f_j = f_j^n$$

and:

$$33) \quad \Delta f_j = f_j^{n+1} - f_j^n$$

then:

$$34) \quad f_j^{n+1} = f_j + \Delta f_j$$

The general implicit finite difference forms are:

1. Time derivative

$$35) \quad \frac{\partial f}{\partial t} \approx \frac{\Delta f}{\Delta t} = \frac{\Delta f_{j+1} + \Delta f_j}{2\Delta t}$$

2. Space derivative

$$36) \quad \frac{\partial f}{\partial x} \approx \frac{\Delta f}{\Delta x} = \frac{(f_{j+1} - f_j) + \theta (\Delta f_{j+1} - \Delta f_j)}{\Delta x}$$

3. Function value

$$37) \quad f \approx \bar{f} = 0.5(f_j + f_{j+1}) + 0.5\theta(\Delta f_j + \Delta f_{j+1})$$

Continuity Equation

The continuity equation describes conservation of mass for the one-dimensional system. From previous text, with the addition of a storage term, S , the continuity equation can be written as

$$38) \quad \frac{\partial A}{\partial t} + \frac{\partial S}{\partial t} + \frac{\partial Q}{\partial x} = q_1$$

Symbol	Description	Units
x	distance along the channel	
t	time	
Q	flow	

Symbol	Description	Units
A	cross-sectional area	
S	storage from non conveying portions of cross section	
q_1	lateral inflow per unit distance	

The above equation can be written for the channel and the floodplain:

$$39) \quad \frac{\partial A_c}{\partial t} + \frac{\partial Q_c}{\partial x_c} = q_f$$

and

$$40) \quad \frac{\partial A_f}{\partial t} + \frac{\partial S}{\partial t} + \frac{\partial Q_f}{\partial x_f} = q_c + q_1$$

where the subscripts c and f refer to the channel and floodplain, respectively, q_1 is the lateral inflow per unit length of floodplain, and q_c and q_f are the exchanges of water between the channel and the floodplain.

NOTE

The HEC-RAS Unsteady flow engine combines the properties of the left and right overbank into a single flow compartment called the floodplain (Finite Difference solution only, not the finite volume solution). Hydraulic properties for the floodplain are computed by combining the left and right overbank elevation vs Area, conveyance, and storage into a single set of relationships for the floodplain portion of the cross section. The reach length used for the floodplain area is computed by taking the arithmetic average of the left and right overbank reach lengths $(L_L + L_R)/2 = L_F$. The average floodplain reach length is used in both the continuity and momentum equations to compute their respective terms for a combined floodplain compartment (Left and right overbank combined together).

This is different than what is done in the Steady Flow computational engine (described above in the previous section), in which the left and right overbank are treated completely separately.

(39) and (40) are now approximated using implicit finite differences by applying (35) through (37):

$$41) \quad \frac{\Delta A_c}{\Delta t} + \frac{\Delta Q_c}{\Delta x_c} = \bar{q}_f$$

$$42) \quad \frac{\Delta A_f}{\Delta t} + \frac{\Delta S}{\Delta t} + \frac{\Delta Q_f}{\Delta x_f} = \bar{q}_c + \bar{q}_1$$

The exchange of mass is equal but not opposite in sign such that $\Delta x_c q_c = -q_f \Delta x_f$.
Adding the above equations together and rearranging yield:

$$43) \quad \Delta Q + \frac{\Delta A_c}{\Delta t} \Delta x_c + \frac{\Delta A_f}{\Delta t} \Delta x_f + \frac{\Delta S}{\Delta t} \Delta x_f = \bar{Q}_t$$

Symbol	Description	Units
$\bar{Q}_t$	the average lateral inflow	
Δx_c	the length of the main channel between two cross sections	
Δx_f	the length of the floodplain between two cross sections (computed as the arithmetic average of the left and right overbank reach lengths)	

Momentum Equation0

The momentum equation states that the rate of change in momentum is equal to the external forces acting on the system. From **Appendix A**, for a single channel:

$$44) \quad \frac{\partial Q}{\partial t} + \frac{\partial(VQ)}{\partial x} + gA \left(\frac{\partial z_s}{\partial x} + S_f \right) = 0$$

The above equation can be written for the channel and for the floodplain:

$$45) \quad \frac{\partial Q_c}{\partial t} + \frac{\partial(V_c Q_c)}{\partial x_c} + gA_c \left(\frac{\partial z_s}{\partial x_c} + S_{fc} \right) = M_f$$

$$46) \quad \frac{\partial Q_f}{\partial t} + \frac{\partial(V_f Q_f)}{\partial x_f} + gA_f \left(\frac{\partial z_s}{\partial x_f} + S_{ff} \right) = M_c$$

where M_c and M_f are the momentum fluxes per unit distance exchanged between the channel and floodplain, respectively. Note that in (45) and (46) the water surface elevation is not subscripted. An assumption in these equations is that the water surface is horizontal at any cross section perpendicular to the flow. Therefore, the water surface elevation is the same for the channel and the floodplain at a given cross section

Using (35) through (37), the above equations are approximated using finite differences:

$$47) \quad \frac{\Delta Q_c}{\Delta t} + \frac{\Delta(V_c Q_c)}{\Delta x_c} + g \bar{A}_c \left(\frac{\Delta z_s}{\Delta x_c} + \bar{S}_{fc} \right) = M_f$$

$$48) \quad \frac{\Delta Q_f}{\Delta t} + \frac{\Delta(V_f Q_f)}{\Delta x_f} + g \bar{A}_f \left(\frac{\Delta z_s}{\Delta x_f} + \bar{S}_{ff} \right) = M_c$$

Note

Note that $\Delta x_c M_c = -\Delta x_f M_f$ (due to the horizontal water surface assumption).

Adding and rearranging the above equations yields:

$$49) \quad \frac{\Delta(Q_c \Delta x_c + Q_f \Delta x_f)}{\Delta t} + \Delta(V_c Q_c) + \Delta(V_f Q_f) + g(A_c + A_f) \Delta z_s + g \bar{A}_c \bar{S}_{fc} \Delta x_c + g \bar{A}_f \bar{S}_{ff} \Delta x_f = 0$$

The final two terms define the friction force from the banks acting on the fluid. An equivalent force can be defined as:

$$50) \quad g \bar{A} \bar{S}_f \Delta x_e = g \bar{A}_c \bar{S}_{fc} \Delta x_c + g \bar{A}_f \bar{S}_{ff} \Delta x_f$$

Now, the convective terms can be rewritten by defining a velocity distribution factor:

$$51) \quad \beta = \frac{V_c^2 A_c + V_f^2 A_f}{V^2 A} = \frac{V_c Q_c + V_f Q_f}{Q V}$$

then:

$$52) \quad \Delta(\beta V Q) = \Delta(V_c Q_c) + \Delta(V_f Q_f)$$

The final form of the momentum equation is:

$$53) \quad \frac{\Delta(Q_c \Delta x_c + Q_f \Delta x_f)}{\Delta t} + \Delta(\beta V Q) + g \bar{A} \Delta z_s + g \bar{A} \bar{S}_f \Delta x_e = 0$$

A more familiar form is obtained by dividing through by Δx_e :

$$54) \quad \frac{\Delta(Q_c \Delta x_c + Q_f \Delta x_f)}{\Delta t \Delta x_e} + \frac{\Delta(\beta V Q)}{\Delta x_e} + g \bar{A} \left(\frac{\Delta z_s}{\Delta x_e} + \bar{S}_f \right) = 0$$

Added Force Term

The friction and pressure forces from the banks do not always describe all the forces that act on the water. Structures such as bridge piers, navigation dams, and cofferdams constrict the flow and exert additional forces, which oppose the flow. In localized areas these forces can predominate and produce a significant increase in water surface elevation (called a "swell head") upstream of the structure.

For a differential distance, dx , the additional forces in the contraction produce a swell head of dh_l . This swell head is only related to the additional forces. The rate of energy loss can be expressed as a local slope:

$$55) \quad S_h = \frac{dh_l}{dx}$$

The friction slope in (54) can be augmented by this term:

$$56) \quad \frac{\partial Q}{\partial t} + \frac{\partial(VQ)}{\partial x} + gA \left(\frac{\partial z_s}{\partial x} + S_f + S_h \right) = 0$$

For steady flow, there are a number of relationships for computation of the swell head upstream of a contraction. For navigation dams, the formulas of Kindsvater and Carter, d'Aubuisson (Chow, 1959), and Nagler were reviewed by Denzel (1961). For bridges, the formulas of Yarnell (WES, 1973) and the Federal Highway Administration (FHWA, 1978) can be used. These formulas were all determined by experimentation and can be expressed in the more general form:

$$57) \quad h_l = C \frac{V^2}{2g}$$

where h_l is the head loss and C is a coefficient. The coefficient C is a function of velocity, depth, and the geometric properties of the opening, but for simplicity, it is assumed to be a constant. The location where the velocity head is evaluated varies from method to method. Generally, the velocity head is evaluated at the tailwater for tranquil flow and at the headwater for supercritical flow in the contraction.

If h_l occurs over a distance Δx_e , then $h_l = \bar{S}_h \Delta x_e$ and $\bar{S}_h = h_l / \Delta x_e$ where $\bar{S}_h$ is the average slope over the interval Δx_e . Within HEC-RAS, the steady flow bridge and culvert routines are used to compute a family of rating curves for the structure. During the simulation, for a given flow and tailwater, a resulting headwater elevation is interpolated from the curves. The difference between the headwater and tailwater is set to h_l and then $\bar{S}_h$ is computed. The result is inserted in the finite difference form of the momentum equation (Equation 2-93), yielding:

$$58) \quad \frac{\Delta(Q_c \Delta x_c + Q_f \Delta x_f)}{\Delta t \Delta x_e} + \frac{\Delta(\beta V Q)}{\Delta x_e} + g \bar{A} \left(\frac{\Delta z_s}{\Delta x_e} + \bar{S}_f + \bar{S}_h \right) = 0$$

Lateral Influx of Momentum

At stream junctions, the momentum as well as the mass of the flow from a tributary enters the receiving stream. If this added momentum is not included in the momentum equation, the entering flow has no momentum and must be accelerated by the flow in the river. The lack of entering momentum causes the convective acceleration term, $\delta(VQ)/\delta x$, to become large. To balance the spatial change in momentum, the water surface slope must be large enough to provide the force to accelerate the fluid. Thus, the water surface has a drop across the reach where the flow enters creating backwater upstream of the junction on the main stem. When the tributary flow is large in relation to that of the receiving stream, the momentum exchange may be significant. The confluence of the Mississippi and Missouri Rivers is such a juncture. During a large flood, the computed decrease in water surface elevation over the Mississippi reach is over 0.5 feet if the influx of momentum is not properly considered.

The entering momentum is given by:

$$59) \quad M_l = \xi \frac{Q_l V_l}{\Delta x}$$

Symbol	Description	Units
Q_l	lateral inflow	
V_l	average velocity of lateral inflow	
ξ	fraction of the momentum entering the receiving stream	

The entering momentum is added to the right side of (58), hence:

$$60) \quad \frac{\Delta(Q_c \Delta x_c + Q_f \Delta x_f)}{\Delta t \Delta x_e} + \frac{\Delta(\beta V Q)}{\Delta x_e} + g \bar{A} \left(\frac{\Delta z_s}{\Delta x_e} + \bar{S}_f + \bar{S}_h \right) = \xi \frac{Q_l V_l}{\Delta x}$$

(60) is only used at stream junctions in a dendritic model.

Finite Difference Form of the Unsteady Flow Equations

(38) and (44) are nonlinear. If the implicit finite difference scheme is directly applied, a system of nonlinear algebraic equations results. Amain and Fang (1970), Fread (1974, 1976) and others have solved the nonlinear equations using the Newton-Raphson iteration technique. Apart from being relatively slow, that iterative scheme can experience troublesome convergence problems at discontinuities in the river geometry. To avoid the nonlinear solution, Preissmann (as reported by Liggett and Cunge, 1975) and Chen (1973) developed a technique

for linearizing the equations. The following section describes how the finite difference equations are linearized in HEC-RAS.

Linearized, Implicit, Finite Difference Equations

The following assumptions are applied:

1. If $f \cdot f \gg \Delta f \cdot \Delta f$, then $\Delta f \cdot \Delta f = 0$ (Preissmann as reported by Liggett and Cunge, 1975).
2. If $g = g(Q, z_s)$, then Δg can be approximated by the first term of the Taylor Series, i.e.:

$$61) \quad \Delta g_j = \left(\frac{\partial g}{\partial Q} \right)_j \Delta Q_j + \left(\frac{\partial g}{\partial z_s} \right)_j \Delta z_j$$

3. If the time step, Δt , is small, then certain variables can be treated explicitly; hence $h_j^{n+1} \approx h_j^n$ and $\Delta h_j \approx 0$.

Assumption 2 is applied to the friction slope, S_f and the area, A . Assumption 3 is applied to the velocity, V , in the convective term; the velocity distribution factor, β ; the equivalent flow path, x ; and the flow distribution factor, ϕ .

The finite difference approximations are listed term by term for the continuity equation in Table 1 and for the momentum equation in Table 2. If the unknown values are grouped on the left-hand side, the following linear equations result:

$$62) \quad CQ1_j \Delta Q_j + CZ1_j \Delta z_{s,j} + CQ2_j \Delta Q_{j+1} + CZ2_j \Delta z_{s,j+1} = CB_j$$

$$63) \quad MQ1_j \Delta Q_j + MZ1_j \Delta z_{s,j} + MQ2_j \Delta Q_{j+1} + MZ2_j \Delta z_{s,j+1} = MB_j$$

Table 1 Finite Difference Approximation of the Terms in the Continuity Equation

Term	Finite Difference Approximation
ΔQ	$(Q_{j+1} - Q_j) + \theta(\Delta Q_{j+1} - Q_j)$
$\frac{\partial A_c}{\partial t} \Delta x_c$	$0.5 \Delta x_{cj} \frac{\left(\frac{dA_c}{dz} \right)_j \Delta z_j + \left(\frac{dA_c}{dz} \right)_{j+1} \Delta z_{j+1}}{\Delta t}$

Term	Finite Difference Approximation
$\frac{\partial A_f}{\partial t} \Delta x_f$	$0.5 \Delta x_{fj} \frac{\left(\frac{dA_f}{dz}\right)_j \Delta z_j + \left(\frac{dA_f}{dz}\right)_{j+1} \Delta z_{j+1}}{\Delta t}$
$\frac{\partial S}{\partial t} \Delta x_f$	$0.5 \Delta x_{fj} \frac{\left(\frac{dS_f}{dz}\right)_j \Delta z_j + \left(\frac{dS_f}{dz}\right)_{j+1} \Delta z_{j+1}}{\Delta t}$

Table 2 Finite Difference Approximation of the Terms in the Momentum Equation

Term	Finite Difference Approximation
$\frac{\partial(Q_c \Delta x_c + Q_f \Delta x_f)}{\partial t \Delta x_e}$	$\frac{0.5}{\Delta x_e \delta t} (\delta Q_{cj} \Delta x_{cj} + \delta Q_{fj} \Delta x_{fj} + \delta Q_{cj+1} \Delta x_{cj} + \delta Q_{fj+1} \Delta x_{fj})$
$\frac{\Delta \beta V Q}{\Delta x_{ej}}$	$\frac{1}{\Delta x_{ej}} [(\beta V Q)_{j+1} - (\beta V Q)_j] + \frac{\theta}{\Delta x_{ej}} [(\beta V Q)_{j+1} - (\beta V Q)_j]$
$g \bar{A} \frac{\Delta z_s}{\Delta x_e}$	$g \bar{A} \left[\frac{z_{s,j+1} - z_{s,j}}{\Delta x_{ej}} + \frac{\theta}{\Delta x_{ej}} (\Delta z_{s,j+1} - \Delta z_{s,j}) \right] + \theta g \Delta \bar{A} \frac{(z_{s,j+1} - z_{s,j})}{\Delta x_{ej}}$
$g \bar{A} (\bar{S}_f + \bar{S}_h)$	$g \bar{A} (\bar{S}_f + \bar{S}_h) + 0.5 \theta g \bar{A} [(\Delta S_{fj+1} + \Delta S_{fj}) + (\Delta S_{hj+1} + \Delta S_{hj})] + 0.5 \theta g (\bar{S}_f + \bar{S}_h) (\Delta A_j + \Delta A_{j+1})$
$\bar{A}$	$0.5 (A_{j+1} + A_j)$
$\bar{S}_f$	$0.5 (S_{fj+1} + S_{fj})$
∂A_j	$\left(\frac{dA}{dZ}\right)_j \Delta z_j$
∂S_{fj}	$\left(\frac{-2S_f}{K} \frac{dK}{dz}\right)_j \Delta z_j + \left(\frac{2s_f}{Q}\right)_j \Delta Q_j$
$\bar{\Delta A}$	$0.5 (\Delta A_j + \Delta A_{j+1})$

The values of the coefficients are defined in Tables 3 and 4.

Table 3 Coefficients for the Continuity Equation

Coefficient	Value
$CQ1_j$	$\frac{-\theta}{\Delta x_{ej}}$
$CZ1_j$	$\frac{0.5}{\Delta t \Delta x_{ej}} \left[\left(\frac{dA_c}{dz} \right)_j \Delta x_{cj} + \left(\frac{dA_f}{dz} + \frac{dS}{dz} \right)_j \Delta x_{fj} \right]$
$CQ2_j$	$\frac{\theta}{\Delta x_{ej}}$
$CZ2_j$	$\frac{0.5}{\Delta t \Delta x_{ej}} \left[\left(\frac{dA_c}{dz} \right)_{j+1} \Delta x_{cj} + \left(\frac{dA_f}{dz} + \frac{dS}{dz} \right)_{j+1} \Delta x_{fj} \right]$
CB_j	$-\frac{Q_{j+1} - Q_j}{\Delta x_{ej}} + \frac{Q_j}{\Delta x_{ej}}$

Table 4 Coefficients of the Momentum Equation

Term	Value
$MQ1_j$	$0.5 \frac{\Delta x_{cj} \phi_j + \Delta x_{fj} (1 - \phi_j)}{\Delta x_{ej} \Delta t} - \frac{\beta_j V_j \theta}{\Delta x_{ej}} + \theta g \bar{A} \frac{(S_{fj} + S_{hj})}{Q_j}$
$MZ1_j$	$\frac{-g \bar{A} \theta}{\Delta x_{ej}} + 0.5 g (z_{j+1} - z_j) \left(\frac{dA}{dz} \right)_j \left(\frac{\theta}{\Delta x_{ej}} \right) - g \theta \bar{A} \left[\left(\frac{dK}{dz} \right)_j \left(\frac{S_{fj}}{K_j} \right) + \left(\frac{dA}{dz} \right)_j \left(\frac{S_{hj}}{A_j} \right) \right] + 0.5 \theta g \left(\frac{dA}{dz} \right)_j (\bar{S}_f + \bar{S}_h)$
$MQ2_j$	$0.5 \left[\Delta x_{cj} \phi_j + 1 + \Delta x_{fj} (1 - \phi_{j+1}) \right] \left(\frac{1}{\Delta x_{ej} \Delta t} \right) + \beta_{j+1} V_{j+1} \left(\frac{\theta}{\Delta x_{ej}} \right) + \frac{\theta g \bar{A}}{Q_{j+1}} (S_{fj+1} + S_{hj+1})$
$MZ2_j$	$\frac{g \bar{A} \theta}{\Delta x_{ej}} + 0.5 g (z_{j+1} - z_j) \left(\frac{dA}{dz} \right)_{j+1} \left(\frac{\theta}{\Delta x_{ej}} \right) - \theta g \bar{A} \left[\left(\frac{dK}{dz} \right)_{j+1} \left(\frac{S_{fj+1}}{K_{j+1}} \right) + \left(\frac{dA}{dz} \right)_{j+1} \left(\frac{S_{hj+1}}{A_{j+1}} \right) \right] + 0.5 \theta g \left(\frac{dA}{dz} \right)_{j+1} (\bar{S}_f + \bar{S}_h)$
MB_j	$-\left[(\beta_{j+1} V_{j+1} Q_{j+1} - \beta_j V_j Q_j) \left(\frac{1}{\Delta x_{ej}} \right) + \left(\frac{g \bar{A}}{\Delta x_{ej}} \right) (z_{j+1} - z_j) + g \bar{A} (\bar{S}_f + \bar{S}_h) \right]$

Flow Distribution Factor

The distribution of flow between the channel and floodplain must be determined. The portion of the flow in the channel is given by:

$$64) \quad \phi_j = \frac{Q_{cj}}{Q_{cj} + Q_{fj}}$$

Fread (1976) assumed that the friction slope is the same for the channel and floodplain, thus the distribution is given by the ratio of conveyance:

$$65) \quad \phi_j = \frac{K_{cj}}{K_{cj} + K_{fj}}$$

(65) is used in the HEC-RAS model.

Equivalent Flow Path

The equivalent flow path is given by:

$$66) \quad \Delta x_e = \frac{\bar{A}_c \bar{S}_{fc} \Delta x_c + \bar{A}_f \bar{S}_{ff} \Delta x_f}{\bar{A}_f \bar{S}_f}$$

If we assume:

$$67) \quad \bar{\phi} = \frac{\bar{K}_c}{\bar{K}_c + \bar{K}_f}$$

where $\bar{\phi}$ is the average flow distribution for the reach, then:

$$68) \quad \Delta x_e = \frac{\bar{A}_c \Delta x_c + \bar{A}_f \Delta x_f}{\bar{A}_f}$$

Since Δx_e is defined explicitly:

$$69) \quad \Delta x_{ej} = \frac{(A_{cj} + A_{cj+1})\Delta x_{cj} + (A_{fj} + A_{fj+1})\Delta x_{fj}}{A_j + A_{j+1}}$$

Boundary Conditions

For a reach of river there are N computational nodes which bound $N-1$ finite difference cells. From these cells $2N-2$ finite difference equations can be developed. Because there are $2N$ unknowns (ΔQ and Δz for each node), two additional equations are needed. These equations are provided by the boundary conditions for each reach, which for subcritical flow, are required at the upstream and downstream ends.

Interior Boundary Conditions (for Reach Connections)

A network is composed of a set of M individual reaches. Interior boundary equations are required to specify connections between reaches. Depending on the type of reach junction, one of two equations is used:

Continuity of flow:

$$70) \quad \sum_{i=1}^l S_{gi} Q_i = 0$$

Symbol	Description	Units
l	the number of reaches connected at a junction	
S_{gi}	-1 if i is a connection to an upstream reach, +1 if i is a connection to a downstream reach	
Q_i	discharge in reach i	

The finite differences form of (70) is:

$$71) \quad \sum_{i=1}^{l-1} MU_{mi} \Delta Q_i + MUQ_m \Delta Q_K = MUB_m$$

Symbol	Description	Units
MU_{mi}	θS_{gi}	
MUQ_m	θS_{gK}	

Continuity of stage:

$$72) \quad z_k = z_c$$

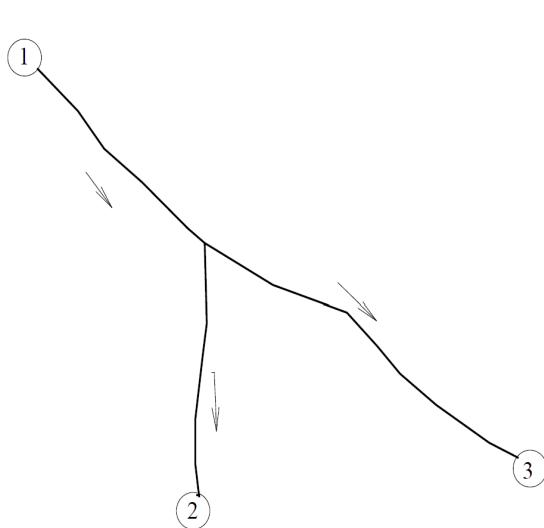
Symbol	Description	Units
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z_k the stage at the boundary of reach k , is set equal to z_c , a stage common to all stage boundary conditions at the junction of interest

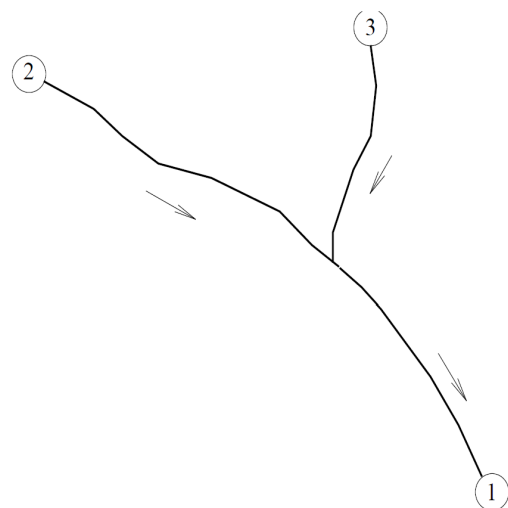
The finite difference form of (72) is:

$$73) \quad MU Z_m \Delta Z_k - MU_m \Delta z_c = MUB_m$$

Symbol	Description	Units
$MU Z_m$	0	
MU_m	0	
MUB_m	$z_c - z_k$	



Flow Split



Flow Combination

Typical flow split and combination.

With reference to the figure above, HEC-RAS uses the following strategy to apply the reach connection boundary condition equations:

- Apply flow continuity to reaches upstream of flow splits and downstream of flow combinations (reach 1 in the figure above). Only one flow boundary equation is used per junction.
- Apply stage continuity for all other reaches (reaches 2 and 3 in the figure above). z_c is computed as the stage corresponding to the flow in reach 1. Therefore, stage in reaches 2 and 3 will be set equal to z_c .

Upstream Boundary Conditions

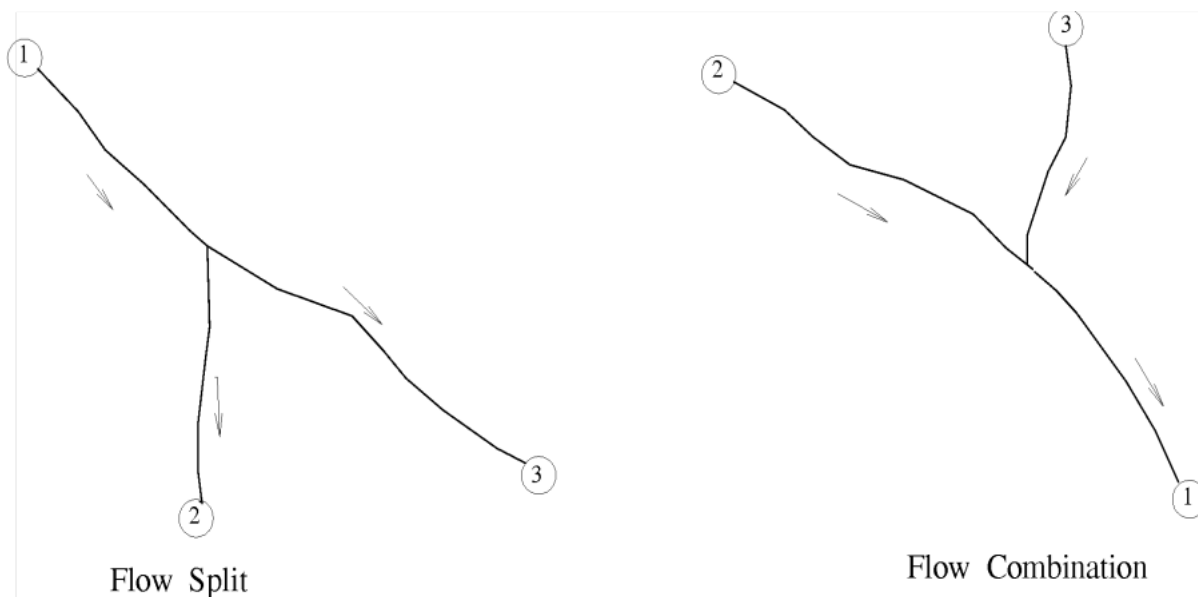
Upstream boundary conditions are required at the upstream end of all reaches that are not connected to other reaches or storage areas. An upstream boundary condition is applied as a flow hydrograph of discharge versus time. The equation of a flow hydrograph for reach m is:

$$74) \quad \Delta Q_k^{n+1} = Q_k^n - Q_k$$

where k is the upstream node of reach m . The finite difference form of (74) is:

$$75) \quad MUQ_m \Delta dQ_K = MUB_m$$

Symbol	Description
MUQ_m	1
MUB_m	$Q_1^{n+1} - Q_1^n$



Typical flow split and combination.

Downstream Boundary Conditions

Downstream boundary conditions are required at the downstream end of all reaches which are not connected to other reaches or storage areas. Four types of downstream boundary conditions can be specified:

- a stage hydrograph,
- a flow hydrograph,
- a single-valued rating curve,
- Normal Depth from Manning's equation.

Stage Hydrograph. A stage hydrograph of water surface elevation versus time may be used as the downstream boundary condition if the stream flows into a backwater environment such as an estuary or bay where the water surface elevation is governed by tidal fluctuations, or where it flows into a lake or reservoir of known stage(s). At time step $(n + 1)\Delta t$, the boundary condition from the stage hydrograph is given by:

$$76) \quad \Delta Z_N = z_N^{n+1} - z_N^n$$

The finite difference form of (76) is:

$$77) \quad CDZ_m \Delta Z_N = CDB_m$$

Symbol	Description
CDZ_m	1
CDB_m	$z_N^{n+1} - z_N^n$

Flow Hydrograph. A flow hydrograph may be used as the downstream boundary condition if recorded gage data is available and the model is being calibrated to a specific flood event. At time step $(n + 1)\Delta t$, the boundary condition from the flow hydrograph is given by the finite difference equation:

$$78) \quad CDQ_m \Delta Q_N = CDB_m$$

Symbol	Description
CDQ_m	1
CDB_m	$Q_N^{n+1} - Q_N^n$

Single Valued Rating Curve. The single valued rating curve is a monotonic function of stage and flow. An example of this type of curve is the steady, uniform flow rating curve. The single valued rating curve can be used to accurately describe the stage-flow relationship of free outfalls such as waterfalls, or hydraulic control structures such as spillways, weirs or lock and dam operations. When applying this type of boundary condition to a natural stream, caution should be used. If the stream location would normally have a looped rating curve, then placing a single valued rating curve as the boundary condition can introduce errors in the solution. To reduce errors in stage, move the boundary condition downstream from your study area, such that it no longer affects the stages in the study area. Further advice is given in (USACE, 1993).

At time $(n + 1)\Delta t$ the boundary condition is given by:

$$79) \quad Q_N + \theta \Delta Q_N = D_{k-1} + \frac{D_k - D_{k-1}}{S_k - S_{k-1}} (z_N + \Delta z_N - S_{k-1})$$

Symbol	Description
D_k	K^{th} discharge ordinate
S_k	K^{th} stage ordinate

After collecting unknown terms on the left side of the equation, the finite difference form of Equation 2-118 is:

$$80) \quad CDQ_m \Delta Q + CDZ_m \Delta z = CDB_m$$

Symbol	Description
CDQ_m	θ
CDZ_m	$\frac{D_k - D_{k-1}}{S_k - S_{k-1}}$
CDB_m	$Q_N + D_{k-1} + \frac{D_k - D_{k-1}}{S_k - S_{k-1}}(z_N - S_{k-1})$

Normal Depth. Use of Manning's equation with a user entered friction slope produces a stage considered to be normal depth if uniform flow conditions existed. Because uniform flow conditions do not normally exist in natural streams, this boundary condition should be used far enough downstream from your study area that it does not affect the results in the study area. Manning's equation may be written as:

$$81) \quad Q = K(S_f)^{0.5}$$

where: K represents the conveyance and S_f is the friction slope.

Skyline Solution of a Sparse System of Linear Equations

The finite difference equations along with external and internal boundary conditions and storage area equations result in a system of linear equations which must be solved for each time step:

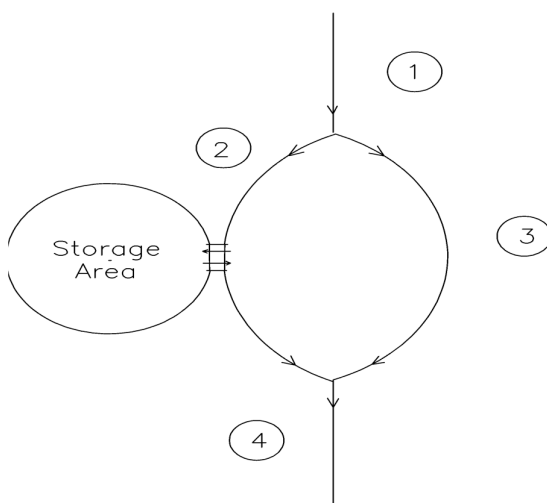
$$82) \quad Ax = b$$

Symbol	Description	Units
A	coefficient matrix	
x	column vector of unknowns	

Symbol	Description	Units
b	column vector of constants	

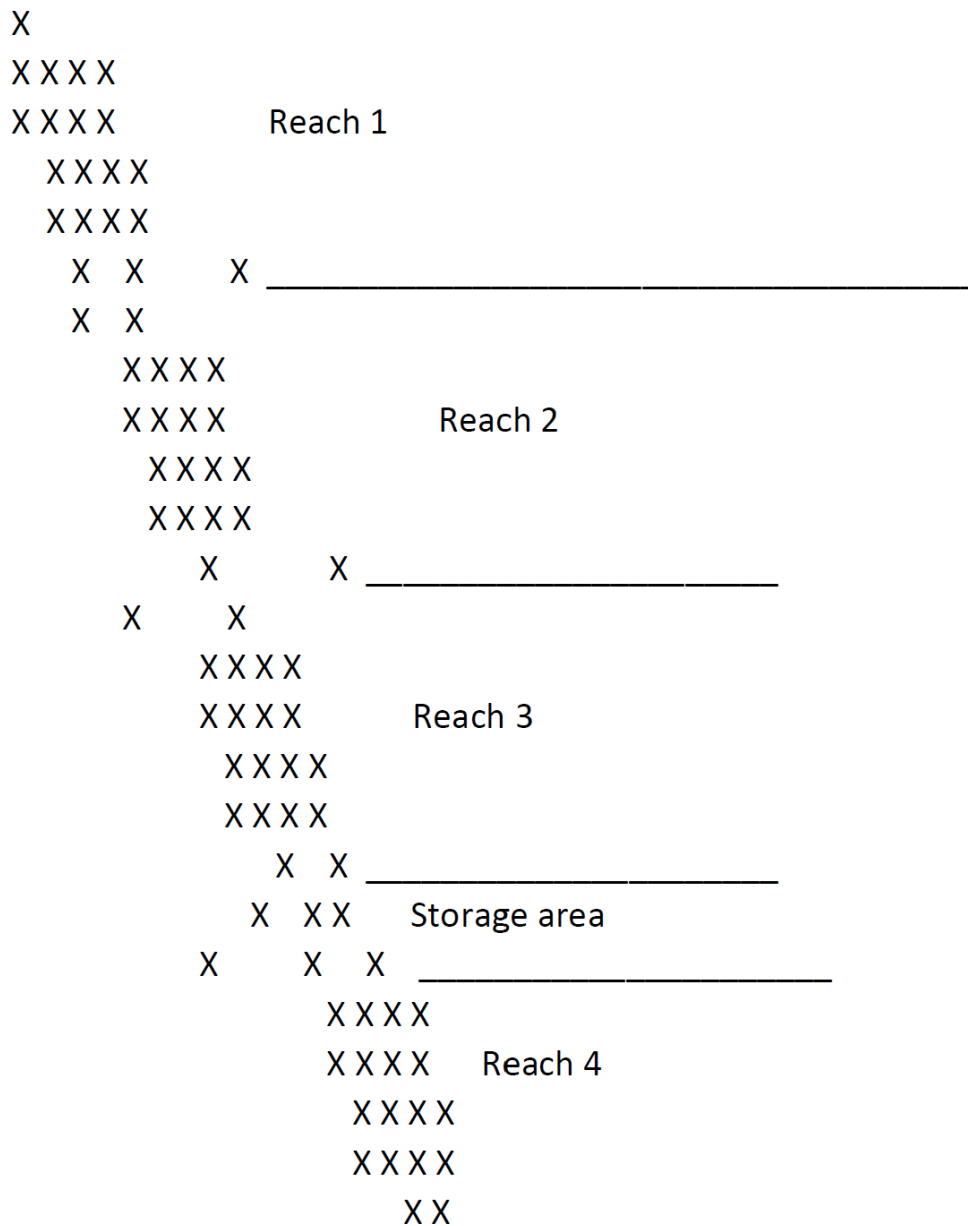
For a single channel without a storage area, the coefficient matrix has a band width of five and can be solved by one of many banded matrix solvers.

For network problems, sparse terms destroy the banded structure. The sparse terms enter and leave at the boundary equations and at the storage areas. The figure below shows a simple system with four reaches and a storage area off of reach 2.



Simple network with four reaches and a storage area.

The corresponding coefficient matrix is shown in the figure below. The elements are banded for the reaches but sparse elements appear at the reach boundaries and at the storage area. This small system is a trivial problem to solve, but systems with hundreds of cross sections and tens of reaches pose a major numerical problem because of the sparse terms. Even the largest computers cannot store the coefficient matrix for a moderately sized problem, furthermore, the computer time required to solve such a large matrix using Gaussian elimination would be very large. Because most of the elements are zero, a majority of computer time would be wasted.



Three practical solution schemes have been used to solve the sparse system of linear equations: Barkau (1985) used a front solver scheme to eliminate terms to the left of the diagonal and pointers to identify sparse columns to the right of the diagonal. Cunge et al. (1980) and Shaffranek (1981) used recursive schemes to significantly reduce the size of the sparse coefficient matrix. Tucci (1978) and Chen and Simons (1979) used the skyline storage scheme (Bathe and Wilson, 1976) to store the coefficient matrix. The goal of these schemes is to more effectively store the coefficient matrix. The front solver and skyline methods identify and store only the significant elements. The recursive schemes are more elegant, significantly reducing the number of linear equations. All use Gaussian elimination to solve the simultaneous equations.

A front solver performs the reduction pass of Gauss elimination before equations are entered into a coefficient matrix. Hence, the coefficient matrix is upper triangular. To further reduce

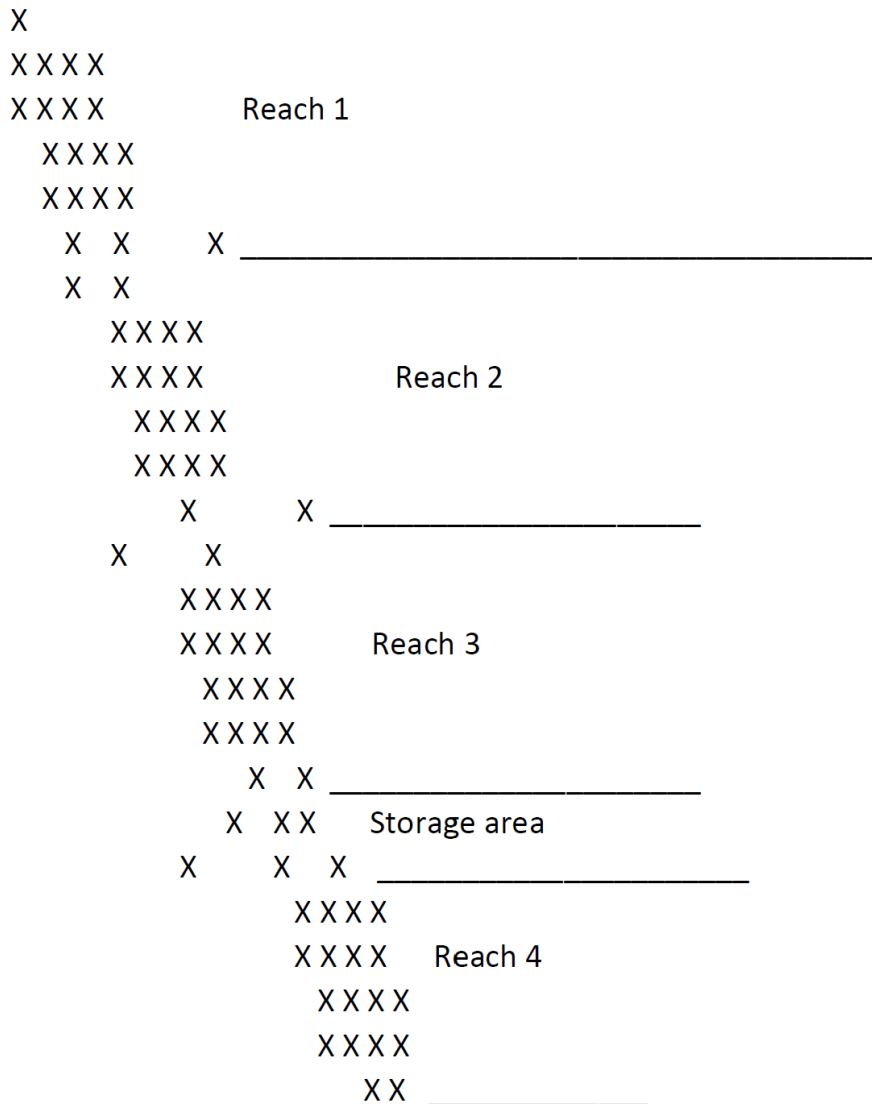
storage, Barkau (1985) proposed indexing sparse columns to the right of the band, thus, only the band and the sparse terms were stored. Since row and column operations were minimized, the procedure should be as fast if not faster than any of the other procedures. But, the procedure could not be readily adapted to a wide variety of problems because of the way that the sparse terms were indexed. Hence, the program needed to be re-dimensioned and recompiled for each new problem.

The recursive schemes are ingenious. Cunge credits the initial application to Friazinov (1970). Cunge's scheme and Schaffranek's scheme are similar in approach but differ greatly in efficiency. Through recursive upward and downward passes, each single routing reach is transformed into two transfer equations which relate the stages and flows at the upstream and downstream boundaries. Cunge substitutes the transfer equations in which M is the number of junctions. Schaffranek combines the transfer equations with the boundary equations, resulting in a system of $4N$ equations in which N is the number of individual reaches. The coefficient matrix is sparse, but the degree is much less than the original system.

By using recursion, the algorithms minimize row and column operations. The key to the algorithm's speed is the solution of a reduced linear equation set. For smaller problems Gaussian elimination on the full matrix would suffice. For larger problems, some type of sparse matrix solver must be used, primarily to reduce the number of elementary operations. Consider, for example, a system of 50 reaches. Schaffranek's matrix would be 200 X 200 and Cunge's matrix would be 50 X 50, 2.7 million and 42,000 operations respectively (the number of operations is approximately $1/3 n^3$ where n is the number of rows).

Another disadvantage of the recursive scheme is adaptability. Lateral weirs which discharge into storage areas or which discharge into other reaches disrupt the recursion algorithm. These weirs may span a short distance or they may span an entire reach. The recursion algorithm, as presented in the above references, will not work for this problem. The algorithm can be adapted, but no documentation has yet been published.

Skyline is the name of a storage algorithm for a sparse matrix. In any sparse matrix, the non-zero elements from the linear system and from the Gaussian elimination procedure are to the left of the diagonal and in a column above the diagonal. This structure is shown in the figure below. Skyline stores these inverted "L shaped" structures in a vector, keeping the total storage at a minimum. Elements in skyline storage are accessed by row and column numbers. Elements outside the "L" are returned as zero, hence the skyline matrix functions exactly as the original matrix. Skyline storage can be adapted to any problem.



The efficiency of Gaussian elimination depends on the number of pointers into skyline storage. Tucci (1978) and Chen and Simons (1979) used the original algorithm as proposed by Bathe and Wilson (1976). This algorithm used only two pointers, the left limit and the upper limit of the "L", thus, a large number of unnecessary elementary operations are performed on zero elements and in searching for rows to reduce. Their solution was acceptable for small problems, but clearly deficient for large problems. Using additional pointers reduces the number of superfluous calculations. If the pointers identify all the sparse columns to the right of the diagonal, then the number of operations is minimized and the performance is similar to the front solver algorithm.

Skyline Solution Algorithm

The skyline storage algorithm was chosen to store the coefficient matrix. The Gauss elimination algorithm of Bathe and Wilson was abandoned because of its poor efficiency. Instead a modified algorithm with seven pointers was developed. The pointers are:

1. IDIA(IROW) - index of the diagonal element in row IROW in skyline storage.

2. ILEFT(IROW) - number of columns to the left of the diagonal.
3. IHIGH(IROW) - number of rows above the diagonal.
4. IRIGHT(IROW) - number of columns in the principal band to the right of the diagonal.
5. ISPCOL(J,IROW) - pointer to sparse columns to the right of the principal band.
6. IZSA(IS) - the row number of storage area IS.
7. IROWZ(N) - the row number of the continuity equation for segment N.

The pointers eliminate the meaningless operations on zero elements. This code is specifically designed for flood routing through a full network.

Computational Procedure

The solution of the water surface elevation at all cross sections, storage areas, and 2D Flow Area cells follows this computational procedure:

1. The solver makes an initial trial at the water surface, flows, derivatives etc... The unsteady flow equations are solved in the implicit finite difference matrix solver (we use a solver called the Skyline Matrix solver) for the 1D nodes. A 2D implicit finite volume solution algorithm is used for the 2D flow areas (See the [2D Theory section](#)).
2. All computational nodes (cross sections, storage areas, and now 2D cells) are checked to see if the computed water surface minus the previous values are less than the numerical solution tolerance.
3. If the error is less than the numerical solution tolerance, then it is finished for that time step; it uses those answers as the correct solution for the time step, and moves on to the next time step.
4. If the numerical error is greater than the tolerance at any node, it iterates, meaning it makes a new estimate of all the derivatives and solves the equations again.
5. During the iteration process, if it comes up with a solution in which the numerical error is less than the tolerance at all locations, it is done and it uses that iteration as the correct answers, and goes on to the next time step.
6. During the iteration (and even first trial) process, the program saves the trial with the least amount of numerical error as being the best solution so far. All water surfaces and flows are saved at all locations.
7. Any iteration that produces a better answer, but does not meet the tolerance, is saved as the current best solution.
8. If the solution goes to the maximum number of iterations (20 by default), then it prints out a warning. However it uses the trial/iteration that had the best answer. It also prints out the location that had the greatest amount of numerical error and the magnitude of that error.
9. This happens even if one of the trials/iterations causes the matrix to go completely unstable. It still does this process and often can find a trial that is not unstable, but does not produce an error less than the numerical tolerance, so it goes with that iteration and moves on.

5 SEMI-IMPLICIT FINITE-VOLUME SCHEME

Hydraulic Equations (1D FV)

Mass Conservation

The continuity equation describing the water volume conservation in 1D is given by

$$83) \quad \frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = q_l$$

where

t = time [T]

Q is the flow [L^3/T]

A is the cross-sectional area [L^2]

q_l is the lateral inflow per unit length [L^2/T]

Momentum Conservation

The momentum equation describing the conservation of momentum in 1D is written as:

$$84) \quad \frac{\partial V}{\partial t} + V \frac{\partial V}{\partial x} + g \left(\frac{\partial H}{\partial x} + S_f + S_h \right) = \nu_t \frac{\partial^2 V}{\partial x^2} + \frac{\tau_{s,x}}{\rho h}$$

where

V = cross-sectionally averaged velocity [L/T]

z_s = water surface elevation [L]

g = gravitational acceleration [L/T^2]

ν_t = turbulent eddy viscosity [L^2/T]

S_f = friction slope [-]

S_h = added force term [-]

τ_s = wind surface stress [$M/L/T^2$]

ρ = water density [M/L^3]

h = water depth [L]

The above equation can be written for the channel and left and right floodplains as (ignoring the wind stresses):

$$85) \quad \frac{\partial V_i}{\partial t} + V_i \frac{\partial V_i}{\partial x} + g \left(\frac{\partial z_s}{\partial x} + S_f + S_h \right) = v_{t,i} \frac{\partial^2 V_i}{\partial x^2} + \frac{\tau_{s,x}}{\rho h_i} + M_i, \quad i \in \{ch, rob, lob\}$$

where *ch*, *lob*, *rob* indicate the channel, and left and right overbanks, respectively. M_i represents the momentum exchange with neighboring cross-sectional areas.

Bottom Friction

Bottom friction represents the energy loss due to skin and form drag on the bed and any other sources of drag including vegetation. The friction slope in the 1D momentum equation is given by

$$86) \quad S_f = \sqrt{\frac{Q}{K}} = \frac{n^2}{R^{4/3}} |V| V$$

where

n = Manning's roughness coefficient [$s/m^{1/3}$]

R = hydraulic radius [L]

V = current velocity [L/T]

Q = discharge [L^3/T]

K = conveyance [L^3/T]

A = cross-sectional area [L^2]

The bottom shear stress is given by

$$87) \quad \tau_b = \gamma R S_f = \rho C_D |V| V = \rho R c_f V$$

where

$\gamma = \rho g$ = water unit weight [$M/L^2/T^2$]

ρ = water density [M/L^3]

C_D = drag coefficient [-]

c_f = friction coefficient [$1/T$]

V = current velocity [L/T]

The drag coefficient, C_D , is related to the nonlinear friction coefficient, c_f , by

$$88) \quad c_f = \frac{C_D}{R} |V|$$

The bottom shear velocity is given by

$$89) \quad u_* = \sqrt{\tau_b / \rho}$$

Eddy Viscosity

Turbulence is a complex phenomenon of chaotic (turbulent) fluid motion and eddies spanning a wide range of length scales. Many of the length scales are too small to be feasibly resolved by a discrete numerical model, so turbulent flow mixing is modeled as a gradient diffusion process. The eddy viscosity is computed as follows,

$$90) \quad \nu_t = D u_* h$$

where D is the mixing coefficient, u_* is the shear velocity, and h is the water depth.

Wind Shear Stress

The wind surface stress vector is calculated as:

$$91) \quad \tau_s = \rho_a C_d |W_{10}| W_{10}$$

where ρ_a is the air density at sea level ($\sim 1.29 \text{ kg/m}^3$), C_d is the wind drag coefficient, W_{10} is the 10-m height wind velocity. The wind speed is calculated using either an Eulerian or Lagrangian reference frame as:

$$92) \quad W_{10} = W_{10}^E - \gamma_W V$$

in which W_{10}^E is the 10-m wind velocity relative to the solid earth (Eulerian wind speed), and γ_W is equal to zero for the Eulerian reference frame or one for the Lagrangian reference frame.

Numerical Methods (1D FV)

Discrete Scheme for SWE. The SWE express volume and momentum conservation. The continuity equation is discretized using finite volume approximations. For the momentum equation, the type of discretization will vary depending on the term. The Crank-Nicolson method is also used to weight the contribution of variables at time steps n and $n+1$. However, the different nature of the equations will call for the use of a more elaborate solver scheme.

Mass Conservation

The continuity equation representing water mass (and volume) conservation is discretized as

$$93) \quad \frac{\Omega_i^{n+1} - \Omega_i^n}{\Delta t} - V_k^{n+\theta} A_k^{n+\theta} + V_{k+1}^{n+\theta} A_{k+1}^{n+\theta} = Q_i$$

where Δt is the time step, Ω_i is the volume at cell i , V_k is the cross-section velocity, A_k is the cross-sectional area, and Q_i represents the cell sources and sinks. The cross-sectional velocity and area are computed here with θ -averaging. Here k and $k + 1$ represents the upstream and downstream cross-sections of cell i , respectively.

Momentum Conservation

Since velocities are computed on cross-sections, the momentum equations are not located on a computational cell. The discrete equations are built based on a semi-implicit scheme in which only the acceleration, barotropic pressure gradient and bottom friction terms contain variables for which the equation is solved. Other terms of the momentum equation are still computed based on the θ -method, but their contribution is smaller and so they are considered explicit forcing function terms and moved to the right-hand side of the linearized system.

Acceleration. Acceleration terms are discretized using a Lagrangian approach as:

$$94) \quad \left(\frac{\partial V_j}{\partial t} + V_j \frac{\partial V_j}{\partial x} \right)_k = \left(\frac{DV_j}{Dt} \right)_k = \frac{V_{k,j}^{n+1} - V_{X(j)}^n}{\Delta t}, \quad j \in \{ch, lob, rob\}$$

where $V_{i,j}^{n+1}$ is the current velocity and $V_{X(j)}^n$ is evaluated at a location $X(j)$. This location is found by integrating the velocity field back in time starting from the location of the computational face. Location $X(i)$ does not in general correspond to a cross-section, so an interpolation technique is applied.

Integration of the velocity is done in steps using the interpolated velocity field in each cell. In practice, this is equivalent to subdividing the integration time step into smaller sub-steps with a Courant number of one or less and increasing the robustness of the computation. In contrast to the explicit Eulerian framework, the semi-Lagrangian scheme allows for the use of large time steps without limiting the stability and with a much-reduced artificial diffusion (regarded as the interpolation error).

Barotropic Pressure Gradient. Recall that the momentum equation is computed at faces, but the water surface elevation term is computed at cells. This staggered grid makes the barotropic pressure gradient ideal for utilizing the simple two-point stencil described previously. In addition, the treated semi-implicitly as

$$95) \quad \left(g \frac{\partial z_s}{\partial x_i} \right)_k \approx g \frac{z_{s,i}^{n+\theta} - z_{s,i-1}^{n+\theta}}{\Delta x_{k,j}}, \quad j \in \{ch, lob, rob\}$$

where g is the gravitational acceleration, $z_s^{n+\theta} = (1 - \theta)z_s^n + \theta z_s^{n+1}$, and $\Delta x_{k,i}$ is the distance between the neighboring cells along i . As a consequence of the implicit weighting, the barotropic pressure gradient consists of two parts, an implicit term with weight θ and an explicit term with weight $(1 - \theta)$.

Momentum Diffusion. Momentum diffusion represents the sum of molecular and turbulence mixing, and momentum dispersion. The momentum diffusion formulation based on the Laplacian of the current velocity is discretized as

$$96) \quad \left(v_{t,j} \frac{\partial^2 V_j}{\partial x^2} \right)_k \approx v_{t,k,j} \left(\frac{\partial^2 V_j}{\partial x^2} \right)_X^n, \quad j \in \{ch, lob, rob\}$$

where $v_{t,j}$ is the explicit face eddy viscosity, and $\left(\frac{\partial^2 V_j}{\partial x^2} \right)_X$ is the Laplacian at location X . The Laplacian is computed at nodes and spatially interpolated at the location X obtained from the acceleration advection. The Laplacian field is explicit in the numerical solution so it will depend only on values computed for the previous time step. The Laplacian terms are calculated using a standard finite-volume approach. Since the velocity is known at the faces, the gradients can be computed for the cells by a simple application of the Gauss' Divergence Theorem on the grid cells. Once the gradients are known for the cells, the Gauss' Divergence theorem is applied again on the dual-grid to obtain a velocity Laplacian at the faces. The face velocity Laplacian at the nodes is computed with a simple inverse distance weighting of the neighboring faces value. Once the Laplacian of the velocity field is known at faces and nodes, the Laplacian term is spatially interpolated using generalized barycentric coordinates to obtain $\left(\frac{\partial^2 V_j}{\partial x^2} \right)_X$. The location X is the same as in the acceleration term.

Bottom Friction. Bottom friction term is computed semi-implicitly in terms of the bottom friction coefficient, as

$$97) \quad (gS_{f,j})_k = c_{f,k,j}^{n+\theta} V_{k,j}^n + 1, \quad j \in \{ch, lob, rob\}$$

Bottom friction is computed semi-implicitly where a bottom friction coefficient, c_f , is computed based on $\bar{V}$ -averaged hydraulic radius and velocities. The bottom friction coefficient is therefore updated during the iteration process. At each of those iterations, a new bottom friction term – $c_f V^{n+1}$ is computed similarly to other implicit terms. The velocity V used in the bottom friction formula is completely implicit for stability purposes.

Solution Procedure. Multiplying the momentum equations for the channel and floodplains by the local area and then summing leads to:

$$98) \quad \sum_{j \in \{ch, lob, rob\}} \frac{A_{k,j}}{A_k} \left[\frac{V_{k,j}^{n+1} - V_{X(j)}^n}{\Delta t} + g \left(\frac{z_{s,i}^{n+\theta} - z_{s,i-1}^{n+\theta}}{\Delta x_{k,j}} + S_{h,j} \right) + c_{f,k,j}^{n+\theta} V_{k,j}^{n+1} - v_{t,k,j}^n \left(\frac{\partial^2 V_j}{\partial x^2} \right)_X^n - \frac{\tau_{s,j}}{\rho h_j} - M_j \right]$$

Each term in the above equation is expanded and approximated as

$$\begin{aligned} \sum_{j \in \{ch, lob, rob\}} \frac{A_{k,j}}{A_k} V_{k,j} &= V_k \\ \sum_{j \in \{ch, lob, rob\}} \frac{A_{k,j}}{A_k} \left(\frac{V_{k,j} - V_{X(j)}}{\Delta t} \right) &= \frac{V_k - \bar{V}_X}{\Delta t} \\ \bar{V}_X &\approx V_X \end{aligned}$$

X = backtracking location corresponding to the velocity V

$$\sum_{j \in \{ch, lob, rob\}} \frac{z_{s,i}^{n+\theta} - z_{s,i-1}^{n+\theta}}{\Delta x_{k,j}} = \frac{z_{s,i}^{n+\theta} - z_{s,i-1}^{n+\theta}}{\Delta x_{k,e}}$$

$X(j)$ = backtracking location corresponding to the velocity V_j

$$\sum_{j \in \{ch, lob, rob\}} c_{f,k,j} \approx c_{f,k} = g \frac{A_k^2}{K_k^2} |V_k|$$

$$\sum_{j \in \{ch, lob, rob\}} v_{t,k,j}^n \left(\frac{\partial^2 V_j}{\partial x^2} \right)_X^n \approx v_{t,k}^n \left(\frac{\partial^2 V}{\partial x^2} \right)_X^n$$

$$\sum_{j \in \{ch, lob, rob\}} \frac{\tau_{s,j}}{\rho h_j} \approx \frac{\tau_s}{\rho h}$$

The final semi-discrete 1D momentum equation is given by

$$99) \quad \frac{V_k^{n+1} - V_X^n}{\Delta t} + g \left(\frac{z_{s,i}^{n+\theta} - z_{s,i-1}^{n+\theta}}{\Delta x_{k,e}} + S_{h_k} \right) + c_{f,k}^{n+\theta} V_k^{n+1} - v_{t,k}^n \left(\frac{\partial^2 V}{\partial x^2} \right)_X^n - \frac{\tau_{s,k}}{\rho h_k}$$

The approximations made in the backtracking velocity and turbulent mixing term are done for efficiency. The approach avoids computing separate backtracking and interpolations of the velocities and velocity Laplacians for the channel and left and right floodplains.

The momentum equation above can be rearranged to obtain an expression for the velocity at as

$$100) \quad V_k^{n+1} = \frac{\Delta t \theta g}{1 + \Delta t c_{f,k}^{n+\theta}} \frac{z_{s,i}^{n+1} - z_{s,i-1}^{n+1}}{\Delta x_e} + F_k$$

where

$$F_k = \frac{B_k^n}{1 + \Delta t c_{f,k}^{n+\theta}}$$

$$B_k^n = V_X^n - (1 - \theta) \Delta t g \frac{z_{s,i}^n - z_{s,i-1}^n}{\Delta x_e} - \Delta t g S_{h,k} + \Delta t v_{t,k}^n \left(\frac{\partial^2 V}{\partial x^2} \right)_X^n + \Delta t \frac{\tau_{s,k}}{\rho h_k^n}$$

To obtain a discrete implicit equation for the water volume, the above equation is inserted into the discrete continuity equation to obtain

$$101) \quad \Omega_i^{n+1} + a_{i,i-1} z_{s,i-1}^{n+1} + a_{i,i} z_{s,i}^{n+1} + a_{i,i+1} z_{s,i+1}^{n+1} = b_i$$

where

$$\begin{aligned}
a_{i,i-1} &= -\frac{\Delta t^2 \theta^2 g A_k^{n+\theta}}{(1+\Delta t c_{f,k}^{n+\theta}) \Delta x_{e,k}} \\
a_{i,i+1} &= -\frac{\Delta t^2 \theta^2 g A_{k+1}^{n+\theta}}{(1+\Delta t c_{f,k+1}^{n+\theta}) \Delta x_{e,k+1}} \\
a_{i,i} &= -a_{i,i-1} - a_{i,i+1} \\
b_i &= \Omega_i^n + \Delta t Q_i + \Delta t \theta (F_k A_k^{n+\theta} - F_{k+1} A_{k+1}^{n+\theta}) + \Delta t (1 - \theta) (V_k^n A_k^{n+\theta} - V_{k+1}^n A_{k+1}^{n+\theta})
\end{aligned}$$

There is an equation of this form for every cell in the domain. Before proceeding, the system of equation for all cells is written in a more compact vector notation.

$$102) \quad \mathbf{\Omega}(\mathbf{Z}) + \mathbf{\Psi} \mathbf{Z} = \mathbf{b}$$

Where $\mathbf{\Omega}$ is the vector of all cell volumes, $\mathbf{Z}$ is the vector of all cell water surface elevations, $\mathbf{\Psi}$ is the coefficient matrix of the system and $\mathbf{b}$ is the right-hand-side vector.

If the coefficients are lagged, the system of equations is mildly non-linear due to the bathymetric relationship for $\mathbf{\Omega}$ as a function of $\mathbf{Z}$. The Jacobian (derivative) of $\mathbf{\Omega}$ with respect to H is given by another bathymetric relationship : the diagonal matrix of cell wet surface areas. If this information is known, a Newton-like technique can be applied to solve the system of equations, producing the iterative formula,

$$103) \quad \mathbf{Z}^{m+1} = \mathbf{Z}^m - [\mathbf{\Psi} + \mathbf{A}^m]^{-1} (\mathbf{\Omega}^m + \mathbf{Z}^m - \mathbf{b})$$

where m denotes the iteration index (not to be confused with the time-step).

Robustness and Stability. When there are no fluxes into a cell the water depth is zero, so the water surface elevation is identical to the previous step. In particular, dry cells remain dry until water flows into them.

In the momentum equation, as water depth decreases to zero, all forces tend to zero. However, the bottom friction is the dominant force, so velocities also go to zero in the limit. As a consequence, it is consistent to assume that dry cells have a flow velocity of zero. The momentum equation for dry faces becomes $\partial V / \partial t = 0$ and dry faces continue to have zero velocity until water flows into them.

As seen in previous sections, both mass and momentum equations are non-linear. Similarly to the DSW solver, an iterative process must be applied. This idea is presented in steps 6 through 9 below.

The Lagrangian treatment of the of acceleration and mixing terms has the advantage of avoiding stability criteria based on the Courant number and mixing.

Discrete Boundary Conditions. Flow boundary conditions are also discretized:

- **Water surface elevation:** The water surface elevation boundary condition is directly implemented as $z_s^{n+1} = z_{s,b}$. The internal cell is then discretized as described above and the terms containing the boundary water surfaces are placed on the right-hand-side of the system of equations.
- **Normal Depth:** The energy grade slope is specified and utilized to compute a flow at each computation face as $Q^{n+1} = K^n S_{f,b}^{1/2}$. Boundary face flows are included in the internal cells as a source term on the right-hand-side of the system of equations.
- **Flow:** The flow boundary condition is specified at each computational face based on the local conveyance. Boundary face flows are included in the internal cells as a source term on the right-hand-side of the system of equations.

Solution Algorithm. The complete solution algorithm is given here:

1. The geometry, local orthogonality and subgrid bathymetry data is obtained or pre-computed.
2. Solution starts with z_s^0 and V_k^0 as the provided initial condition at time step $n=0$.
3. Boundary conditions are provided for the next time step $n+1$.
4. Initial guess $z_s^{n+1} = z_s^n$ and $V_k^{n+1} = V_k^n$.
5. Compute explicit terms that remain constant through the time step compute, such as the mixing term and wind forcing.
6. Compute θ -averaged water surface elevation and subgrid bathymetry quantities that depend on the computed elevation (face areas, fluid surface areas, hydraulic radii, Manning's n , etc.).
7. A system of equations is assembled for the continuity at the cells as described in the previous sections.
8. The system of equations is solved iteratively using the Newton-like algorithm with the given boundary conditions to obtain a candidate solution z_s^{n+1} .
9. Velocities V_k^{n+1} are computed based on the discrete momentum equation.
10. If the residual (or alternatively, the correction) is larger than a given tolerance (and the maximum number of iterations has not been reached), go back to step 6; otherwise, continue with the next step.
11. The computed solution is accepted.
12. Increment n . If there are more time steps go back to step 3, otherwise end.

The loop provided by steps 6 through 10 has the purpose of updating the coefficients of the system of equations, so that the solution of the nonlinear system (rather than its linearization) is obtained at every time step. As expected, a fully nonlinear solution has very desirable properties such as wetting several cells and propagating waves through several cells in a single time step.